

	NEUROLEPTICS (FGA)		ATYPICAL ANTIPSYCHOTICS (SGA)		
Drug Classes	<u>Phenothiazine</u>	<u>Butyrophenone</u>	“peens”	“dones”	“pip” and “rip”
	- Chlorpromazine - Fluphenazine	haloperidol	Clozapine (Clozaril) Olanzapine (Zyprexa)	Risperidone (Risperdal)	Aripiprazole (Abilify)
Drug Actions	- Hi affinity dopamine D2 antagonist - H1, α1-AR, M1 antagonist		- Lo affinity dopamine D2 antagonist - hi affinity 5-HT 2A antagonist, or 5-HT 1A partial agonist, or D2/D3 partial agonist		
Drug Effects	<u>Therapeutic effects</u>	<u>Side effects</u>	<u>Therapeutic effects</u>	<u>Side effects</u>	
	D2 antagonism in mesolimbic - antipsychotic - positive > negative (or cog) symptoms	D2 antagonism in nigrostriatal - EPS (treated with benztropine) - TD (treated with valbenzine)	- Antipsychotic - Clozapine may be superior to other atypicals	Clozapine - Agranulocytosis - Constipation - Hypersalivation (paradoxical) More general - Cardiometabolic / Weight gain	
		D2 antagonism in tuberohypophyseal endocrine / hyperprolactinemia		Decreased incidence of EPS / endocrine possibly due to 5-HTergic modulation of DA release in nigrostriatal and tuberohypophyseal DA pathways	
			HAM side effects - H1 – sedation - α1-AR - ↓ bp / hr - M1 – dry mouth, constipation	Atypicals are generally preferred over neuroleptics - favorable side-effect profile - olanzapine, risperdal are often 1st choice All (?) atypicals also FDA approved as monotherapy for bipolar disorder Many atypicals also used as adjunct in treatment of MDD	

1. Broad Spectrum

- low affinity D2 antag

- 5HT_{2A} antag

Clozapine — D4 antag

olanzapine

sedation
H₁ M₃
antag
cardio met.

- ⊕ 5HT_{1A} partial ag

Quetiapine — H₁

lurasidone — 5HT₇

ziprasidone — 5HT_{2C} inverse agonist

↳ No weight gain
Met.

2. D2/D3 partial ag

aripiprazole

brexipiprazole

CATIE ~ 1500 ppl, 18 mo
65% Olanzapine

Phase 1: Perphenazine (active control)
Quetiapine
Risperdone
Ziprasadone

>80%

~ 75%
Discontinue

Phase 2:
Clozapine or Switch
Efficacy

Tolerability

~ 1/2 resp. to Clozapine
20% to anything else
clin. sup.

olanzapine
risperdone
- Stayed on longest